

The Physics of Fluids

Pascal's Principle and Hydraulic Systems

General Science – Grade 11



Hydraulic Jack



Car Brake



Excavator



Barber Chair



Have you ever wondered how pressure applied to one end of a toothpaste tube forces paste out the other?



Why does water flow with such force from a faucet, even if the source is far away?

Motivation

Think-Pair-Share

Have you ever wondered...



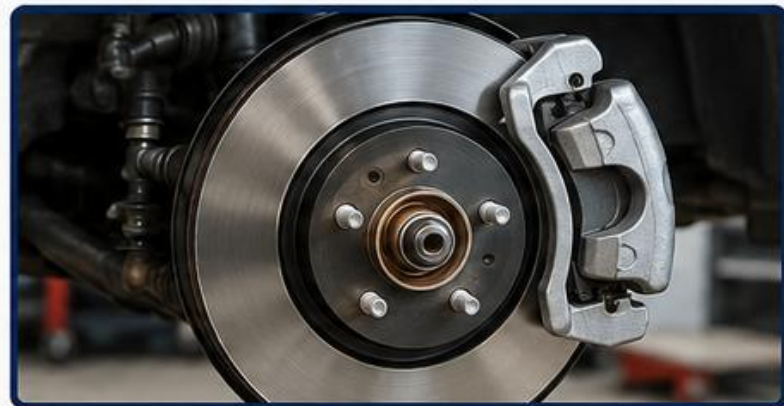
How can a mechanic lift an entire car?



Why does pressing one brake pedal stop all four wheels?



How does a barber chair move up and down so easily?



These technologies all use fluids and **Pascal's Principle!**

What is a Fluid?



Fluid

A fluid is any substance that can **flow** and take the **shape of its container**.

Types of Fluids

Liquids



Gases



Water



Oil



Air



Blood



Key Idea: Fluids have **no fixed shape**.

BLAISE PASCAL (1623-1662)



Blaise Pascal was a French mathematician, physicist, inventor, philosopher, and writer who made significant contributions to mathematics, physics, engineering, and probability. He is best known in physics for formulating **Pascal's Principle**, which explains how pressure is transmitted through fluids.

Major Contributions

1. Pascal's Principle

Pascal discovered that: Pressure applied to a confined fluid is transmitted equally and undiminished in all directions throughout the fluid.

This principle is the basis of hydraulic systems, such as:

Hydraulic brakes

Hydraulic lifts

Hydraulic presses

Car jacks

Heavy construction equipment

2. Pascaline

Pascal invented one of the world's first mechanical calculators, called the Pascaline, to help his father perform tax calculations.

3. Mathematics

He made important contributions to:

Probability theory

Geometry

The study of conic sections

What is Pressure?



Pressure

Pressure is the force applied over a certain area.

Formula

Pressure = Force ÷ Area

$$P = \frac{F}{A}$$

SI Unit

Pascal (Pa)

Remember:

↑ More force → More pressure

➡ Larger area → Less pressure

Small Area (Thumb)



Same force with a small area results in **HIGH PRESSURE**

Large Area (Flat Hand)



Same force with a large area results in **LOW PRESSURE**



Pressure depends on both the **force** applied and the **area** over which it is applied.

FACTORS AFFECTING PRESSURE

FORCE DEPENDENCY



**Direct relationship;
More force=More pressure**

AREA DEPENDENCY



**Inverse relationship;
Smaller area= Higher pressure**

DENSITY

the amount of mass contained in a given volume of a substance. It helps determine whether an object will float or sink in a fluid.

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$\rho = \frac{m}{V}$$

BOUYANCY

the upward force exerted by a fluid
(liquid or gas) on an object immersed in it

What Determines Whether an Object Floats or

Object's Density Compared to the Fluid	What Happens
Less dense than the fluid	Floats
Equal to the fluid	Suspends (neutral buoyancy)
More dense than the fluid	Sinks

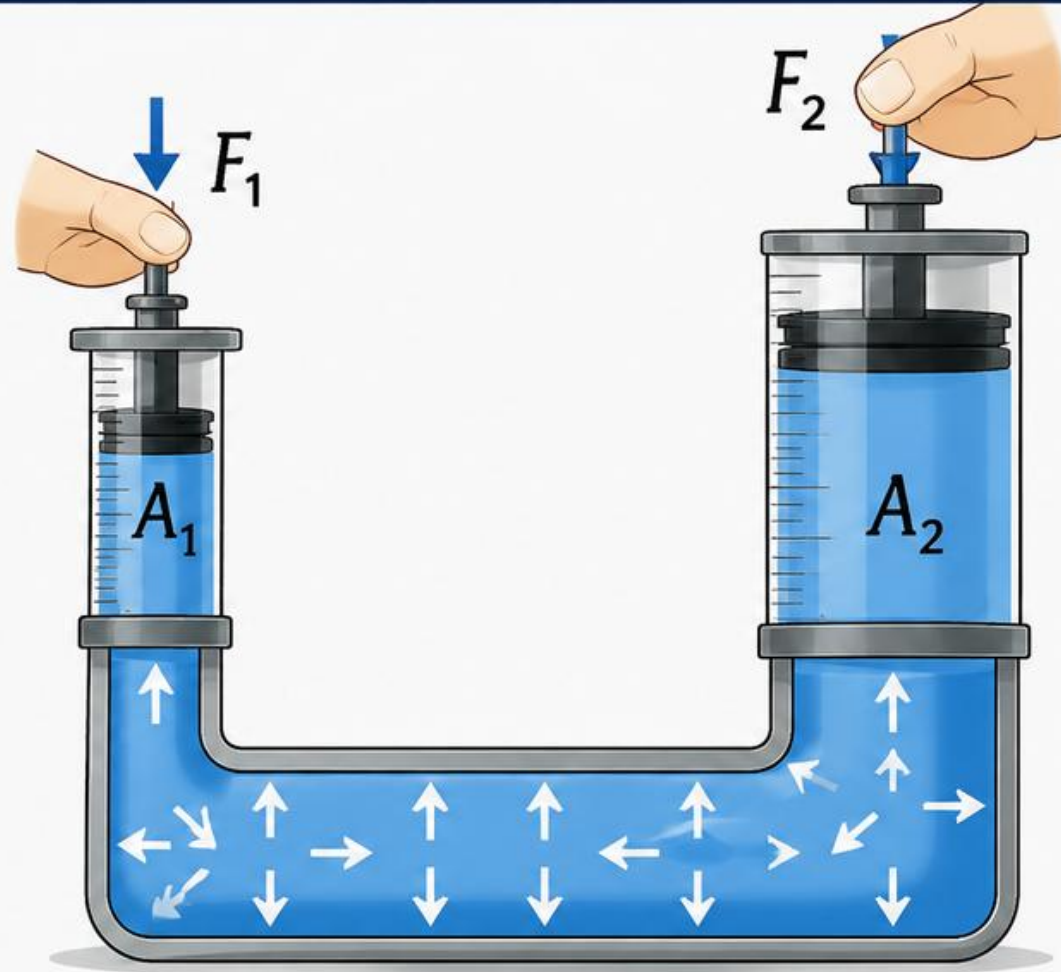
Pascal's Principle

Blaise Pascal discovered that:
Pressure applied to a confined fluid is
transmitted equally in all directions.

In simple words:

Push one part of a sealed liquid...

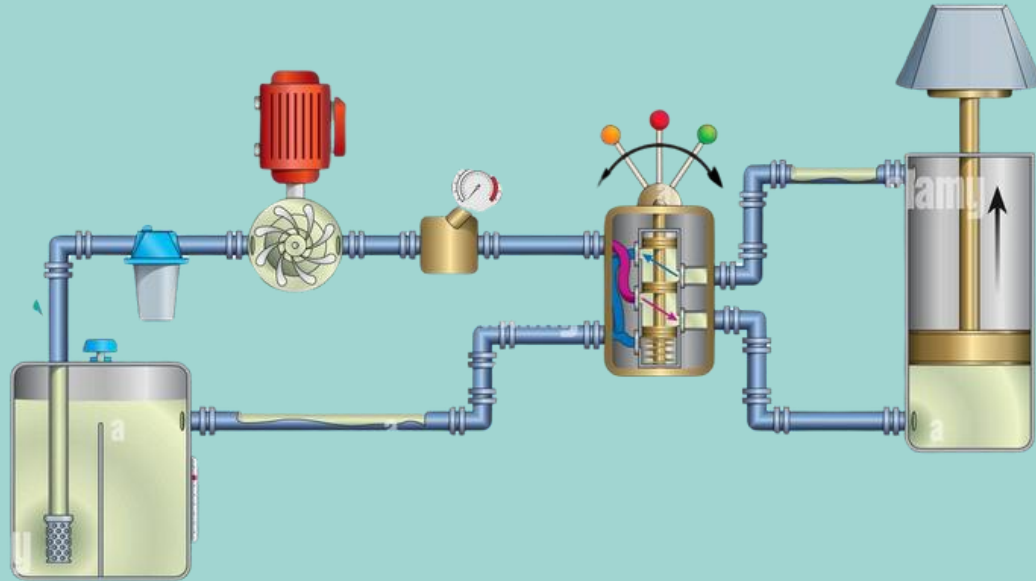
- ➔ Pressure spreads throughout the liquid.
- ➔ Every part experiences the **same pressure.**



Pascal's Principle is the foundation of many hydraulic systems we use in everyday life!

HYDRAULIC SYSTEM

- A hydraulic system is a machine that transfer force from one point to another.

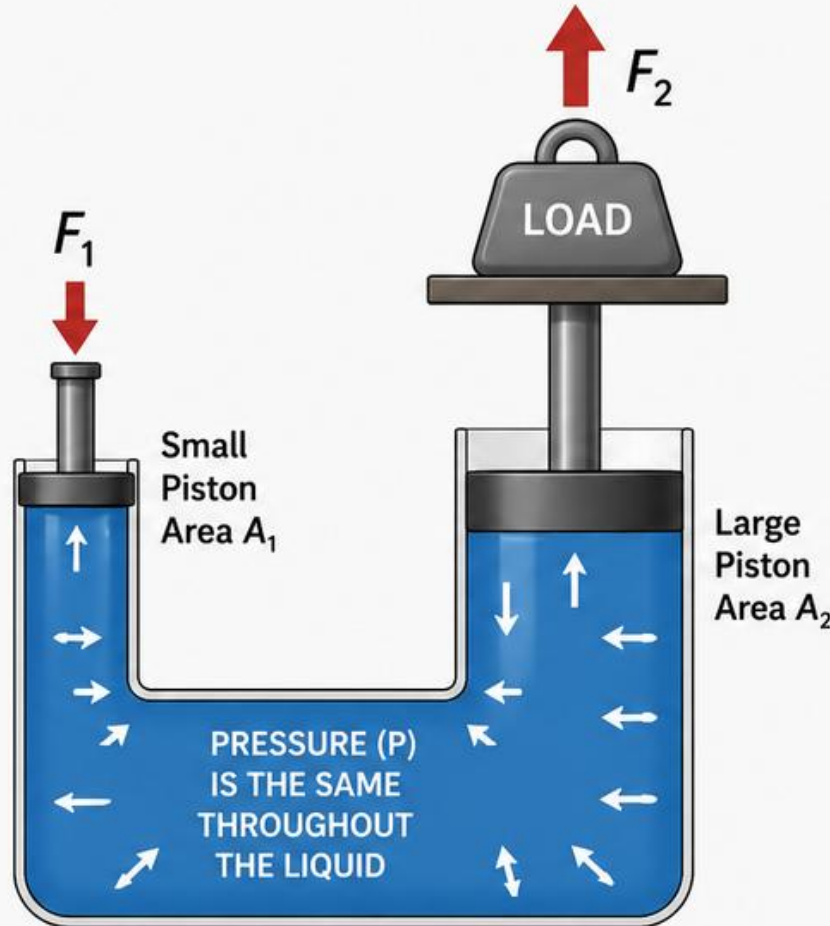


How Hydraulic Systems Work

Hydraulic System

A hydraulic system uses:

- Confined liquid
- Pressure
- Two pistons of different sizes



$$P_1 = P_2 \quad \text{or} \quad \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

What happens?

Small Force



Small Piston



Pressure travels through liquid



Large Piston



Large Force



Result:

Heavy objects can be lifted with little effort.



Sample Problem (Finding output force F_2)

A hydraulic lift has a small piston with an area of 0.02 m^2 . A force of 100 N is applied to it. The large piston has an area of 0.50 m^2 . Find the output force produced by the large piston.

Piston=solid cylinder metal

Given:

- $A_1 = 0.02 \text{ m}^2$
- $F_1 = 100 \text{ N}$
- $A_2 = 0.50 \text{ m}^2$

Solution:

Substitute the given values:

$$\frac{100}{0.02} = \frac{F_2}{0.50}$$

Solve for F_2 :

$$F_2 = \frac{100 \times 0.50}{0.02}$$

$$F_2 = \frac{50}{0.02}$$

$$F_2 = 2500 \text{ N}$$

Formula (Pascal's Principle):

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$F_1 = \text{Input force}$

$F_2 = \text{Output force}$

$A_1 = \text{Area of small piston}$

$A_2 = \text{Area of large piston}$

Sample Problem (Finding output force F_2)

A hydraulic system has a small piston with an area of **0.25 m²** and a large piston with an area of **0.25 m²**. A force of **400 N** is applied to the small piston. What is the output force?

Given:

- $A_1 = 0.25 \text{ m}^2$
- $A_2 = 0.25 \text{ m}^2$
- $F_1 = 400 \text{ N}$

Solution:

Using Pascal's Principle:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

Substitute the values:

$$\frac{400}{0.25} = \frac{F_2}{0.25}$$

Cross multiply:

$$400(0.25) = 0.25F_2$$

$$100 = 0.25F_2$$

Divide both sides by **0.25**:

$$F_2 = 400 \text{ N}$$

Sample Problem (Finding output force F_2)

A hydraulic lift has two pistons, each with an area of **0.10 m²**. If the input force is **250 N**, what is the output force?

Solution

Since the piston areas are equal,

$$F_2 = F_1$$

$$F_2 = 250 \text{ N}$$

Answer:

250 N

Sample Problem Using ($P_1 = P_2$)

A hydraulic lift has a small piston with an area of **0.04 m²** and a large piston with an area of **0.20 m²**. A force of **100 N** is applied to the small piston.

Find the pressure in the system.

Given:

• $F_1 = 100 \text{ N}$

• $A_1 = 0.04 \text{ m}^2$

Step 1: Find the input pressure.

$$P_1 = \frac{F_1}{A_1}$$

$$P_1 = \frac{100}{0.04}$$

$$P_1 = 2,500 \text{ Pa}$$

Step 2: Apply Pascal's Principle.

Since

$$P_1 = P_2$$

then

$$P_2 = 2,500 \text{ Pa}$$

ASSESSMENT

Copy and Answer. Use GUESS method.

1. A hydraulic lift has a small piston with an area of **0.04 m^2** . A force of **120 N** is applied. The large piston has an area of **0.80 m^2** . **Find the output force.**
2. A force of 250 N is applied to a piston with an area of 0.05 m^2 . **Find the pressure.**

Real-Life Applications

Hydraulic Systems Around Us

Machine		Application		
	Hydraulic Jack	→	Lifts vehicles	
	Hydraulic Brake	→	Stops vehicles	
	Hydraulic Lift	→	Raises heavy equipment	
	Excavator	→	Moves soil and rocks	
	Barber Chair	→	Adjusts seat height	
	Dentist Chair	→	Raises and lowers patients	

Why Are Hydraulic Systems Important?

Hydraulic systems help us:



✓ Lift heavy loads



✓ Reduce human effort



✓ Increase safety



✓ Improve efficiency



✓ Save time



Lift heavy loads



Reduce human effort



Increase safety



Improve efficiency



Save time



Think About It



Imagine doing these jobs
without hydraulic systems!

Worked Example 1



Situation

A mechanic uses a hydraulic jack to lift a car.



Question

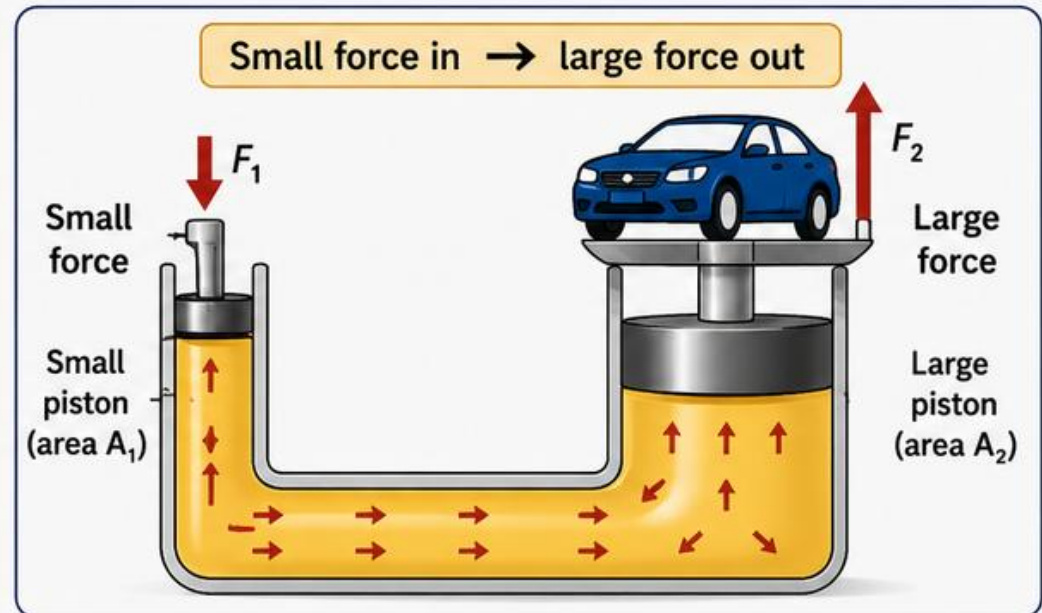
Which scientific principle allows the jack to lift the car?



Answer

Pascal's Principle

Pressure applied to the hydraulic fluid is transmitted equally, allowing a small force to lift a heavy vehicle.



Worked Example 2



Situation

A driver presses the brake pedal.



Question

Why do all four brakes work at the same time?



Answer

Brake fluid transmits pressure equally throughout the braking system according to **Pascal's Principle**.



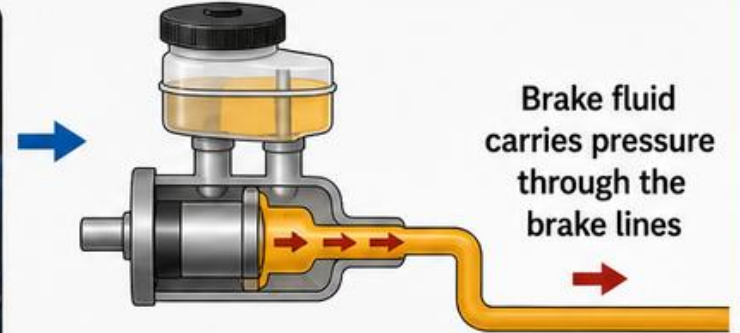
Result:

All four brakes apply force at the same time, stopping the vehicle safely and effectively.

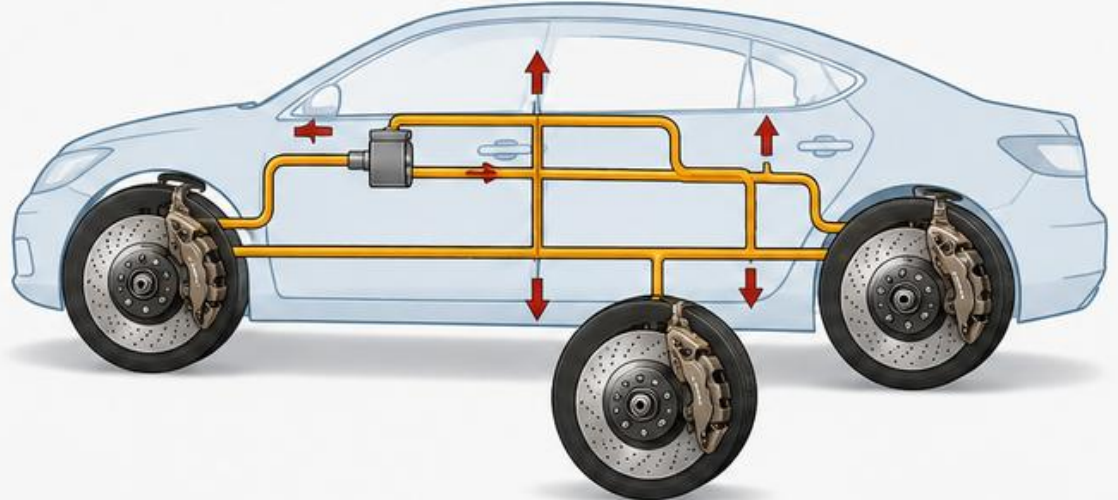
Driver presses
brake pedal



Pressure is created
in the master cylinder



Pressure is transmitted equally
in all directions (**Pascal's Principle**)



Key Takeaways

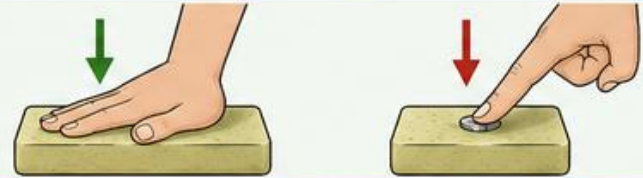
Remember!



Fluids can flow and take the shape of their container.



Pressure is force applied over an area.



Pascal's Principle states that pressure in a confined fluid is transmitted equally.



Hydraulic systems **multiply force**.



Hydraulic systems make work **easier, safer, and more efficient**.



Group Activity

Pressure in Action



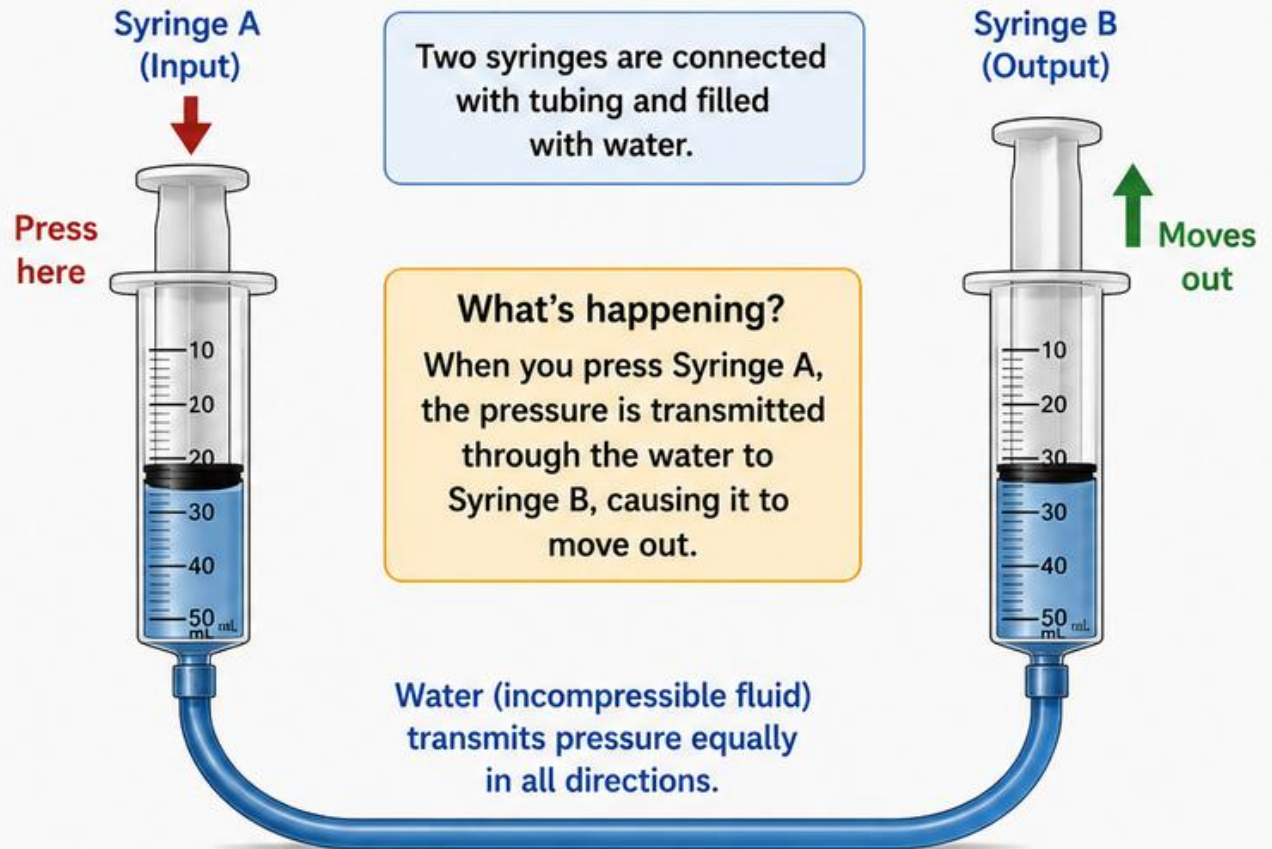
Materials

- 2 Syringes
- Plastic tubing
- Water



Procedure

- 1 Fill both syringes and tubing with water.
- 2 Connect the syringes.
- 3 Press one syringe slowly.
- 4 Observe the other syringe.
- 5 Record your observations.



Conclusion: What did you observe? How does this demonstrate pressure and Pascal's Principle?

Guide Questions

Discuss with your group.

1 What happened when you pushed one syringe?

?

2 Why did the other syringe move?

?

3 What scientific principle explains this activity?

?

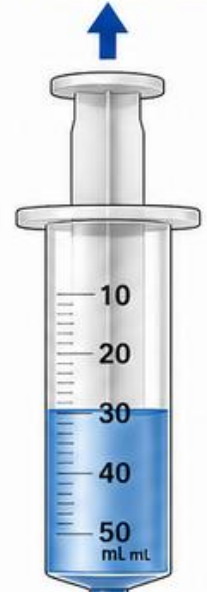
Push here



Two syringes are connected with tubing and filled with water.

When you push one syringe, the pressure is transmitted through the water to the other syringe, causing it to move out.

It moves out



Pressure is transmitted equally throughout the fluid.
This is **Pascal's Principle**.

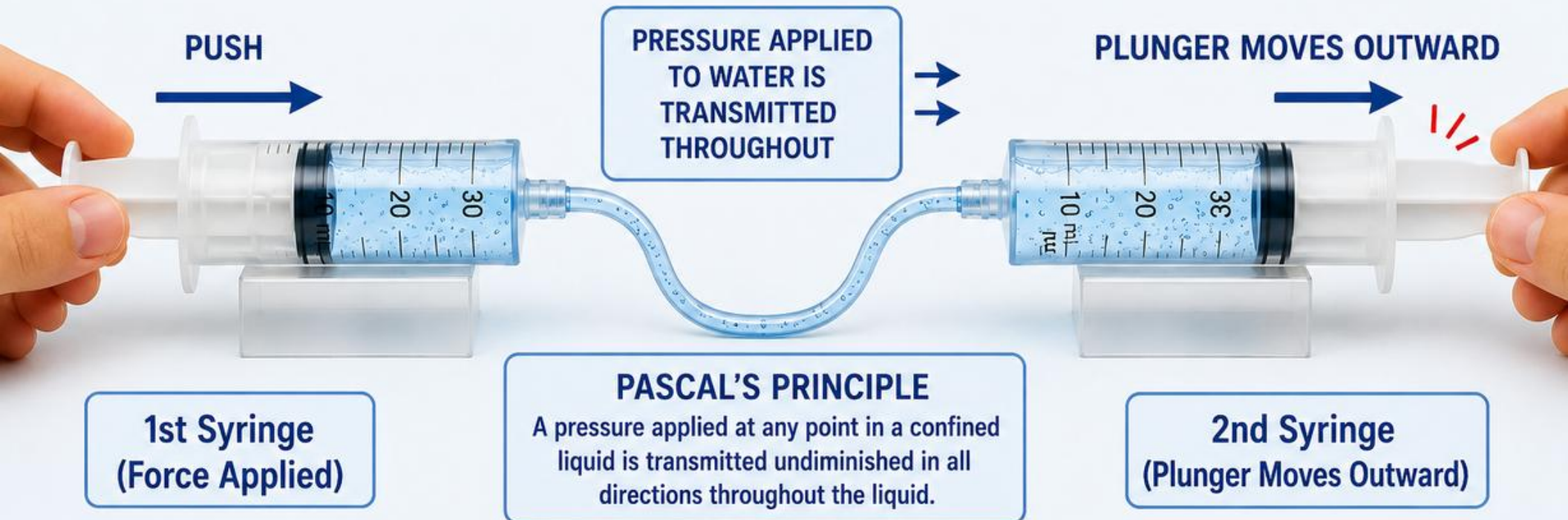
1. What happened when you pushed one syringe?

Answer: When one syringe was pushed, the plunger of the other syringe moved outward. The pressure applied to the water in the first syringe was transmitted through the connected tubing, causing the second syringe to respond.



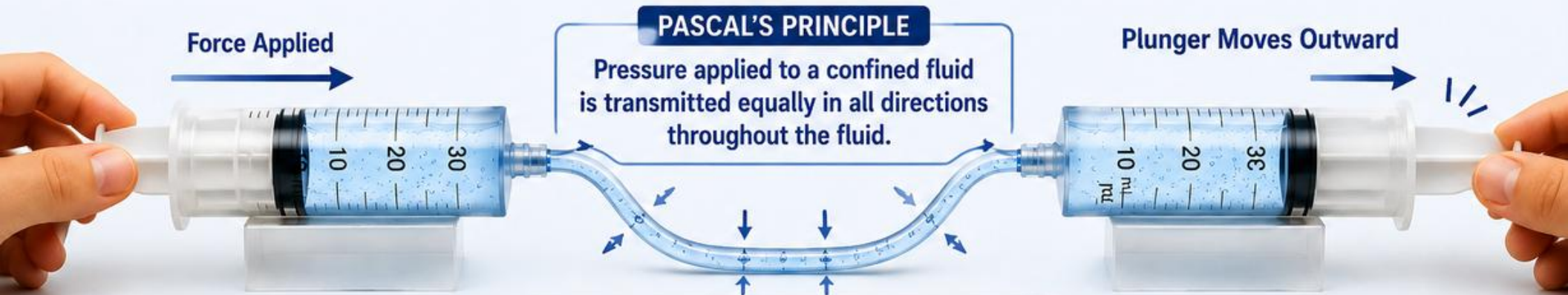
2. Why did the other syringe move?

Answer: The other syringe moved because the pressure applied to the confined water was transmitted throughout the liquid. Since liquids are nearly incompressible, the pressure was transferred to the second syringe, pushing its plunger outward.



3. What principle explains the activity?

Answer: The activity is explained by **Pascal's Principle**, which states that pressure applied to a confined fluid is transmitted equally in all directions throughout the fluid. As a result, the pressure applied to one syringe was transferred through the water to the other syringe, causing it to move. This principle is the basis for the operation of hydraulic systems such as hydraulic jacks, hydraulic brakes, and hydraulic lifts.



APPLICATIONS IN HYDRAULIC SYSTEMS

Hydraulic Jack



A small force applied to the pump is transmitted to a larger area, producing a large lifting force.

Hydraulic Brakes



Pressure applied to the brake fluid is transmitted to the brake calipers, stopping the vehicle.

Hydraulic Lift



Pressure applied to the hydraulic fluid lifts heavy loads with a small input force.